

Conic combinations

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A convex cone is a set containing all conic combinations of its elements, *i.e.*, a set $K \subseteq \mathbb{R}^n$ satisfying $\text{cone}(K) = K$. In this note we present some basic properties of conic combinations. We first show that the conic hull $\text{cone}(S)$ of any set $S \subseteq \mathbb{R}^n$ is a convex cone.

Theorem 1 (*The conic hull of a set is a convex cone*)

For every set $S \subseteq \mathbb{R}^n$, its conic hull $\text{cone}(S)$ is a convex cone. ■

Proof: We must show $\text{cone}(S)$ contains all conic combinations of its elements. Therefore let $d = \mu_1 d^1 + \dots + \mu_k d^k$ be a conic combination of k directions $d^1, \dots, d^k \in \text{cone}(S)$, where $\mu > 0_k$. We must show $d \in \text{cone}(S)$.

Let $i \in \{1, \dots, k\}$ be arbitrary. Since $d^i \in \text{cone}(S)$, the direction d^i is a conic combination of directions in S , *i.e.*, one may write $d^i = \sum_{j \in J_i} \delta_{i,j} \cdot e^{i,j}$, where J_i is a finite index set, $\delta_{i,j} \geq 0$ and $e^{i,j} \in S$ for $j \in J_i$. One then has:

$$d = \mu_1 d^1 + \dots + \mu_k d^k = \sum_{i=1}^k \sum_{j \in J_i} (\mu_i \delta_{i,j}) \cdot e^{i,j},$$

which gives d as a conic combination of points in S . Hence $d \in \text{cone}(S)$. ■

A convex cone is a set $K \subseteq \mathbb{R}^n$ satisfying $\text{cone}(K) = K$. Therefore, since Lemma 1 shows $\text{cone}(S)$ is a convex cone for every $S \subseteq \mathbb{R}^n$, one has:

Corollary 1 (*The conic hull of a conic hull*)

For any set $S \subseteq \mathbb{R}^n$ one has $\text{cone}(\text{cone}(S)) = \text{cone}(S)$. ■

Since directions in a convex cone can be scaled with positive numbers, a direction is a conic combination of non-equivalent directions in K if and only if it is a sum of non-equivalent directions in K :

Observation 1 (*Conic combinations of directions vs. sums of directions*)

A direction in a convex cone K is a conic combination of non-equivalent directions in K $\iff$ it is a sum of non-equivalent directions in K . ■

Indeed, if $d = \mu_1 d^1 + \dots + \mu_k d^k$ holds, where $\mu \geq 0_k$, and none of $d^1, \dots, d^k \in K \setminus \{0_n\}$ are equivalent to d , then one has $d = r^1 + \dots + r^k$, where none of the directions $r^j := \mu_j \cdot d^j$ for $j \in \{1, \dots, k\}$ are equivalent to d .